

HealthCost Lens

Distributed Medicare Analytics Platform

COURSE	DSE-3222: Big Data Analytics and Tools
DEPARTMENT	Data Science and Engineering
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DATA SOURCE

CMS Medicare Provider Utilization and Payment Data (Public Use File)

1. Executive Summary

HealthCost Lens is a distributed big-data analytics platform designed to surface inefficiencies, anomalies, and access gaps within U.S. Medicare fee-for-service billing data. Built as part of DSE-3222: Big Data Analytics and Tools, the system processes publicly available CMS Medicare Provider Utilization and Payment Data through a full distributed pipeline — from raw ingestion to interactive visualisation — enabling stakeholders to explore billing patterns, identify outlier providers, and understand geographic disparities in healthcare reimbursement.

9.8M	6.8 GB	3-Node	5
Provider-Procedure Rows	Total Data Ingested	Hadoop Cluster	Analytical Queries

Core Analytical Capabilities

- **Geographic Payout Analysis:** Average Medicare reimbursement mapped by state and procedure code
- **Anomaly Detection:** Statistical identification of high-billing outlier providers ($|z\text{-score}| > 2$)
- **Access Gap Analysis:** Beneficiary-to-provider ratios revealing underserved regions
- **Cost & Volume Analysis:** Procedure-level payment rate efficiency and specialty spend breakdown
- **ML Intelligence:** Random Forest payment prediction and K-Means provider clustering

Dataset	Cluster	Analytical Queries	ML Models
CMS Medicare PUF	3-Node Hadoop	5 (Q1–Q5)	K-Means + Random Forest

2. Project Background & Objectives

2.1 Context

Medicare is the United States federal health insurance programme serving approximately 65 million Americans aged 65 and older, or with qualifying disabilities. The CMS (Centers for Medicare & Medicaid Services) publishes annual Provider Utilization and Payment Data files as part of its transparency initiative. These Public Use Files (PUFs) contain aggregated claim data at the provider-procedure level, covering all Part B fee-for-service claims across all 50 U.S. states. Despite their public availability, these datasets are large, heterogeneous, and not easily analysed without distributed computing infrastructure.

2.2 Problem Statement

The project addresses four interconnected questions:

- Do Medicare reimbursement rates vary significantly across U.S. states for the same procedure? If so, which states overpay or underpay?
- Are there identifiable providers whose billing behaviour deviates so substantially from peers that they may represent fraud, overutilisation, or data quality issues?
- Which regions of the country are most underserved — where the ratio of Medicare beneficiaries to available providers is highest?
- Can machine learning models predict Medicare payment amounts from provider and procedure features, and which features drive those predictions?

2.3 Objectives

- Build a scalable distributed data pipeline capable of processing the full CMS PUF dataset on a multi-node Hadoop cluster
- Implement five analytical PySpark SQL queries covering geographic payout, anomaly detection, access gaps, cost analysis, and specialty volume
- Apply Spark MLlib to train and evaluate unsupervised (K-Means clustering) and supervised (Random Forest regression) models
- Deliver findings through a fully interactive, production-quality Streamlit dashboard

3. Data Source & Schema

3.1 Source

The primary dataset is the CMS Medicare Provider Utilization and Payment Data — Part B Physician and Other Suppliers Public Use File, available at data.cms.gov. The file covers fee-for-service claims aggregated at the individual provider and HCPCS procedure code level for calendar year 2021.

3.2 Datasets Used

Dataset	Size	Description
Provider & Service 2021	2.9 GB	9.8M rows — core billing table with NPI, HCPCS, charges, payments
Part D Prescriber 2020	3.4 GB	25.2M rows — drug prescriptions by provider
Geography & Service 2020	41 MB	268K rows — state and ZIP-level cost and volume figures
NPPES NPI Registry (synthetic)	34 MB	1.1M rows — provider specialty and entity type lookup

3.3 Key Fields

Field	Type	Description
npi	String	National Provider Identifier — unique 10-digit provider ID
proc_code	String	HCPCS procedure code identifying the service billed
state	String	U.S. state where the provider is located (2-letter code)
specialty	String	Provider specialty designation (e.g., Internal Medicine)
submitted_charge	Float	Average submitted charge per service across all claims
medicare_payment	Float	Average Medicare reimbursement per service
bene_count	Integer	Total distinct beneficiaries served

3.4 Data Characteristics

- **Coverage:** All 50 U.S. states plus Washington D.C. and territories
- **Granularity:** Provider-procedure level — one row per unique NPI + HCPCS combination
- **Format:** CSV on ingestion; tab-separated values (TSV) after ETL processing in HDFS
- **Privacy:** No individual patient data present; CMS suppresses rows with fewer than 11 beneficiaries before publication

4. System Architecture & Pipeline

4.1 Distributed Infrastructure

The platform is built on a three-node Hadoop cluster configured with HDFS for distributed storage and YARN for resource management. This architecture ensures that data processing scales horizontally and that no single node represents a point of failure during analytical query execution. The cluster is deployed using Docker Compose, with each node running as an isolated container on the same bridge network.

Component	Technology	Role	Nodes
Distributed Storage	HDFS	Raw CSV and TSV data storage with replication factor 2	3
Resource Management	YARN	Job scheduling and cluster resource allocation	3
ETL Processing	PySpark	Distributed data cleaning, type casting, and enrichment	All
Query Engine	SparkSQL	SQL-based analytical queries Q1 through Q5	All
ML Framework	Spark MLlib	K-Means clustering and Random Forest regression	All
Dashboard	Streamlit	Interactive web-based visualisation layer	Single host

4.2 Pipeline Stages

Stage 1 — Data Ingestion

Raw CMS CSV files were uploaded from the local filesystem into HDFS using `hdfs dfs -put` commands. Block distribution across both DataNodes was verified using `hdfs fsck` before proceeding.

Stage 2 — Data Preparation (ETL)

PySpark reads the raw CSV files from HDFS, filters suppressed rows, casts string columns to numeric types, normalises provider type names to lowercase, and performs a left outer join with the NPI specialty registry. The enriched dataset of 9,886,177 rows is written back to HDFS at `/medicare/processed/`.

Stage 3 — Analytical Query Execution

Five SparkSQL queries are executed against the processed data registered as a temporary view. Each query result is materialised as a CSV output partition under `/medicare/results/`, with one subdirectory per query.

Stage 4 — Machine Learning

Two Spark MLlib pipelines are trained on the processed dataset. K-Means clusters providers into four behavioural groups. Random Forest regresses average Medicare payment from billing features.

Stage 5 — Dashboard Visualisation

The Streamlit application reads all query result CSVs and serves five interactive pages with Plotly Express charts. No data is cached between sessions; all loading occurs at request time.

5. Analytical Queries (Q1–Q5)

Five analytical queries form the backbone of the platform. Each corresponds to a distinct research question and produces a structured CSV output that feeds a dashboard page. All queries are written in SparkSQL and executed against a temporary view of the processed 9.8-million-row dataset.

Q1 — Geographic Payout Distribution

Output: q1_payment_by_state

Computes the average Medicare payment per procedure code, grouped by state. Ranks the top 30 procedure codes by average payment to focus the analysis on the most financially significant procedures. Drives the choropleth map on the Geography dashboard page.

Key Finding: Maryland pays \$41,215 for procedure Q2043 (immunotherapy), compared to significantly lower rates in other states — a variation exceeding 40%.

Q2 — High-Billing Outlier Detection

Output: q2_outliers

Identifies providers whose submitted charges for a given procedure deviate by more than two standard deviations from the procedure mean ($|z\text{-score}| > 2$). Z-scores are computed per procedure code using window functions across all providers.

Key Finding: NPI 1376645499 in Texas charged \$22,329 for procedure 0011A when the national average was \$45 — a Z-score of 198.

Q3 — Geographic Access Gap

Output: q3_access_gap

Computes the ratio of Medicare beneficiaries to available providers for each state-specialty combination. A high ratio indicates that providers in that region carry a disproportionate caseload, signalling underservice or workforce shortage.

Key Finding: North Carolina Internal Medicine shows 2,694 beneficiaries per provider — nearly five times a healthy ratio.

Q4 — Cost Disparity & Payment Rate Analysis

Output: q4_yoy_trend

Analyses the relationship between provider submitted charges and actual Medicare payments for each procedure. Computes the unpaid gap, surfacing procedures where Medicare reimbursement falls furthest below billed charges.

Key Finding: Procedure C9600 (cardiac catheterisation): providers charge \$35,758 on average but receive only \$4,610 — a payment rate of 12.9%.

Q5 — Specialty Volume & Efficiency

Output: q5_specialty_volume

Aggregates total Medicare payment volume by specialty, along with provider count, total beneficiaries served, and average payment per claim. Identifies the specialties that account for the largest share of Medicare expenditure.

Key Finding: Orthopedic Surgery leads total payment volume at \$8.19 billion across 137 million beneficiaries.

6. Machine Learning Models

6.1 K-Means Provider Clustering

An unsupervised K-Means clustering model is trained using Spark MLlib to group providers into behaviourally similar clusters based on their billing features. The elbow method determines the optimal number of clusters by plotting WSSSE (Within Set Sum of Squared Errors) against k values from 2 to 8. Features are standardised via StandardScaler (mean=0, std=1).

k	WSSSE	Interpretation
2	4,402,633	Two broad groups — insufficient granularity
3	3,105,788	Improving separation
4	2,548,810	Elbow point — optimal cluster count
5	2,213,791	Diminishing returns begin
6	1,891,550	Overfitting risk increases

The four clusters correspond to the following provider archetypes:

- **Cluster 0 — Low-volume routine billers:** General practitioners with moderate charges and few distinct procedures
- **Cluster 1 — High-charge specialist providers:** Small patient volumes but significantly above-average submitted charges
- **Cluster 2 — High-volume primary care:** Large beneficiary counts with near-average charges — typical family practice profile
- **Cluster 3 — Outlier/anomalous billing behaviour:** Extreme submitted charges relative to Medicare payments

6.2 Random Forest Payment Prediction

A supervised Random Forest regressor is trained to predict avg_payment (the Medicare reimbursement amount) from provider and procedure features. Trained on an 80/20 train-test split using Pipeline with StringIndexer stages for categorical variables.

RMSE (Test Set)	R ² (Test Set)	Top Feature	Train/Test Split
\$208.65	0.2781	Submitted Charge (~96% importance)	80 / 20

6.3 Model Interpretation & Improvements

The R² of 0.28 indicates moderate predictive power. The high RMSE (\$208.65) reflects significant billing heterogeneity — procedures range from routine office visits (\$20–50) to complex surgical interventions (\$2,000+). Recommended improvements:

- Engineer specialty-level and geographic features with one-hot encoding
- Train per-procedure sub-models to reduce within-group variance
- Experiment with GBTRegressor for improved non-linear capture
- Apply log-transformation to avg_payment to reduce skew before regression

7. Dashboard Design & Features

7.1 Technology Stack

Layer	Technology
Web Framework	Streamlit (Python)
Charting	Plotly Express (interactive SVG charts)
Data Processing	Pandas (CSV loading and filtering)
Navigation	Session-state button nav with custom CSS
Data Source	Hive/SparkSQL CSV output partitions (part-*.csv)

7.2 Dashboard Pages

Geography

Interactive USA choropleth map coloured by average Medicare payment per state. Filterable by procedure code with metric cards for state count, highest-paying state, and lowest-paying state.

Anomalies

Two-column layout displaying flagged outlier providers (Q2) alongside a grouped bar chart of geographic access gaps (Q3), surfacing billing anomalies and access equity concerns side by side.

Costs

Bubble scatter chart mapping procedure payment rate (%) against average submitted charge, with bubble size encoding the unpaid gap, plus a pie chart of top 10 specialties by total Medicare payment.

ML & Intelligence

Model performance metrics (RMSE, R^2 , primary feature) as metric cards. Bubble chart for specialty efficiency, bar heatmap for anomaly concentration by state, and saved PNG artefacts.

Global Search

Cross-dataset search by NPI or HCPCS procedure code, returning results from Q1, Q2, and Q4 in parallel with contextual inline charts for matching procedure codes.

8. Key Findings

>40% Inter-state payout variation	198 Top outlier Z-score	2,694 Max beneficiaries/provider (NC)	12.9% Lowest reimbursement rate
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8.1 Geographic Disparities

The Q1 analysis reveals substantial inter-state variation in Medicare reimbursement for identical procedure codes. For high-cost procedures, payout differences between the highest and lowest reimbursing states can exceed 40%, reflecting geographic price indices, local cost-of-living adjustments, or differences in provider mix. Maryland consistently appears as a high-reimbursement state for specialty procedures.

8.2 Billing Anomalies

The Z-score based outlier analysis (Q2) identifies a small proportion of providers — typically 2–5% of the dataset — whose submitted charges deviate by more than two standard deviations from procedure means. These providers are concentrated in a subset of states and specialties. The top outlier (NPI 1376645499, Texas, procedure 0011A) recorded a Z-score of 198, charging \$22,329 against a \$45 national average.

8.3 Access Gaps

The access gap analysis (Q3) identifies significant beneficiary-to-provider concentration in several states for specialist procedure categories. North Carolina Internal Medicine shows the most severe gap at 2,694 beneficiaries per provider. Cardiology and Orthopedic Surgery show the most consistently strained access ratios across multiple states.

8.4 Cost Efficiency

The payment rate analysis (Q4) demonstrates that Medicare's effective reimbursement rate varies substantially across procedure types. Procedure C9600 (cardiac catheterisation) is reimbursed at approximately 12.9% of the submitted charge — \$4,610 against a \$35,758 submitted price.

8.5 Specialty Concentration

Orthopedic Surgery leads total payment volume at \$8.19 billion, followed by Internal Medicine (\$8.04 billion) and Radiology (\$7.71 billion). The top seven specialties collectively represent the vast majority of total Part B payment volume, with procedural specialties tending to have higher per-claim averages than primary care specialties.

9. Limitations & Future Work

9.1 Current Limitations

- **Single-year snapshot:** Covers only 2021 CMS data; multi-year trending not yet implemented. The Q4 folder is named `yoy_trend` but reflects a single-period analysis.
- **Top-N windowing:** Q1 and Q4 results are limited to top-30 and top-50 procedure codes respectively. Procedures outside these windows return no results in Global Search.
- **Aggregated data:** CMS suppresses cells with fewer than 11 beneficiaries, meaning rare procedures and small-population providers are systematically underrepresented.
- **ML feature sparsity:** The Random Forest uses only numeric billing features. No geographic or specialty encodings are included, limiting model expressiveness ($R^2 = 0.28$).
- **Static outputs:** Dashboard reads pre-computed CSV files. There is no live query capability; re-running the pipeline is required to reflect updated data.

9.2 Recommended Future Work

- Multi-year ingestion: Ingest 3–5 years of CMS PUF data to enable genuine year-over-year trend analysis and temporal anomaly detection
- Feature engineering: Add one-hot encoded specialty and state features; experiment with procedure category groupings using HCPCS code prefixes
- Per-procedure models: Train individual regression models per HCPCS code to dramatically reduce within-group prediction variance
- Real-time pipeline: Integrate Apache Kafka for streaming ingestion of CMS data releases and trigger incremental Spark jobs
- HBase point lookups: Implement Apache HBase for fast NPI-based provider lookups
- Export functionality: Add PDF report export directly from the dashboard using WeasyPrint or ReportLab

10. Conclusion

HealthCost Lens demonstrates that a full distributed big-data pipeline — from raw CSV ingestion through Hadoop/HDFS, PySpark ETL transformation, SparkSQL analytical queries, Spark MLlib modelling, and interactive Streamlit visualisation — can be assembled as a coherent, production-quality system within an academic project context.

The five analytical queries deliver substantive insights into U.S. Medicare billing patterns: geographic payout disparities exceeding 40% for identical procedures, statistical billing anomalies with Z-scores reaching 198, access gap ratios identifying severely underserved regions, cost efficiency variation showing reimbursement as low as 12.9% of submitted charges, and specialty expenditure concentration revealing Orthopedic Surgery and Internal Medicine as the dominant cost drivers.

The machine learning layer adds predictive and clustering capabilities that extend the purely descriptive analyses. The K-Means model identifies four distinct provider archetypes with actionable policy interpretations. The Random Forest model, while achieving moderate R^2 , confirms that submitted charge is the dominant predictor of Medicare payment — consistent with the programme's prospective payment structure.

The Streamlit dashboard makes these findings accessible to non-technical stakeholders through interactive maps, charts, and a cross-dataset search interface. With multi-year temporal analysis, richer ML feature engineering, and per-procedure sub-model training, HealthCost Lens could serve as a practical tool for Medicare policy research, fraud detection support, and healthcare access equity assessment.

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